

Empirical Scaling of Transformer Architectures for Low-Resource Languages

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Abstract

Recent progress in generative large language models has accelerated translation accuracy, but low-resource languages face data scarcity. This paper investigates parameter-efficient adaptation of transformer models across 12 low-resource language pairs.

1. Introduction

Natural language processing models have demonstrated exceptional capabilities across high-resource languages. However, language disparities persist due to limited parallel corpus availability. We address this bottleneck through sparse cross-attention adaptation.

2. Methodology

We employ a 12-layer transformer encoder-decoder with low-rank adaptation (LoRA) modules. Our evaluation dataset consists of the FLORES-200 benchmark with 10,000 sentence pairs per target language. We measure BLEU and chrF++ scores.

3. Results and Evaluation

The proposed sparse adapter framework achieves 31.4 BLEU, outperforming full fine-tuning baselines by 3.2 BLEU while reducing trainable parameters by 88%. Evaluation shows consistent improvements across all target dialects.

4. Conclusion and Future Scope

We demonstrate that modular parameter adaptation bridges low-resource language translation gaps. Future work includes expanding to zero-shot speech-to-text translation and addressing compute limits.